Clinical Diagnostic Handbook

A systems-based teaching reference for junior doctors

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Scope and clinical governance

This handbook provides a practical introduction to common and high-consequence adult medical conditions. It is not exhaustive, does not replace local emergency protocols or specialty guidelines, and is not a validated point-of-care decision support system. Diagnostic thresholds depend on patient characteristics, assay, clinical setting and guideline jurisdiction. Pediatric, obstetric, neonatal and critical-care pathways require dedicated references.

Clinical priority: assess airway, breathing, circulation, disability and exposure first; stabilize and escalate before pursuing a complete differential. Never allow a normal initial investigation to override a concerning clinical presentation.

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1. Clinical diagnostic method and emergency recognition

Step 1 — Establish acuity

Identify instability, time-critical symptoms, vital-sign abnormalities, altered mental status and rapidly progressive illness.

Step 2 — Define the syndrome

Document symptom onset, chronology, associated symptoms, risk factors, exposures, medications, family history and focused examination.

Step 3 — Prioritize the differential

Separate cannot-miss diagnoses from likely diagnoses; record evidence for and against each possibility.

Step 4 — Choose tests purposefully

Specify whether each test screens, rules out in a defined population, supports, confirms, stages or monitors a condition.

Step 5 — Reassess

Review trends, treatment response, pending results, discordant data and disposition safety.

Time-critical red flags

- Acute chest pain, syncope or shock: urgent ECG and emergency evaluation for ACS, dissection, pulmonary embolism and other critical causes.

- Sudden focal neurological deficit: activate stroke pathway; record last-known-well time; urgent brain imaging.
- Severe breathlessness, cyanosis, inability to speak full sentences or deteriorating oxygenation: emergency airway/respiratory assessment.
- Sepsis concern with hypotension, confusion, tachypnea or organ dysfunction: immediate resuscitation and infection evaluation.
- Severe hyperglycemia with vomiting, deep breathing or confusion: evaluate for DKA/HHS, including ketones, pH/bicarbonate and electrolytes.
- GI hemorrhage, meningism with toxicity, anaphylaxis, severe hyperkalemia or pregnancy-related emergencies: use immediate local protocols.

2. Disease profiles by organ system

Endocrine & metabolic

Type 1 diabetes — Mechanism: Autoimmune beta-cell destruction. Presentation: Polyuria, polydipsia, weight loss; DKA possible.

Investigations: Plasma glucose, HbA1c; islet antibodies, C-peptide. **Interpretation / caution:** Diabetes criteria establish hyperglycemia; antibodies and C-peptide support classification; assess ketones and acidosis when unwell.

Type 2 diabetes — Mechanism: Insulin resistance with beta-cell dysfunction. Presentation: Often silent; polyuria, thirst, blurred vision.

Investigations: FPG, HbA1c, 75-g OGTT. **Interpretation / caution:** FPG >=126 mg/dL, HbA1c >=6.5%, or 2-h OGTT >=200 mg/dL; generally confirm absent unequivocal hyperglycemia.

Prediabetes — Mechanism: Impaired glycemic regulation. Presentation: Usually asymptomatic.

Investigations: FPG, HbA1c, OGTT. **Interpretation / caution:** ADA ranges: FPG 100–125 mg/dL; HbA1c 5.7–6.4%; 2-h OGTT 140–199 mg/dL.

Primary hypothyroidism — Mechanism: Often autoimmune thyroiditis. Presentation: Fatigue, constipation, cold intolerance, bradycardia.

Investigations: TSH, free T4; anti-TPO if appropriate. **Interpretation / caution:** High TSH + low FT4 supports overt primary hypothyroidism; high TSH + normal FT4 suggests subclinical disease.

Thyrotoxicosis — Mechanism: Graves disease, toxic nodules or thyroiditis. Presentation: Tremor, weight loss, heat intolerance, palpitations.

Investigations: TSH, FT4, T3; TRAb, imaging when appropriate. **Interpretation / caution:** Suppressed TSH + elevated thyroid hormones supports overt thyrotoxicosis; establish etiology before therapy.

Adrenal insufficiency — Mechanism: Adrenal or pituitary dysfunction; steroid withdrawal. Presentation: Weight loss, weakness, hypotension, nausea.

Investigations: Morning cortisol, ACTH, stimulation test, electrolytes. **Interpretation / caution:** Interpret by collection time and assay; suspected adrenal crisis is an emergency and testing must not delay care.

Cushing syndrome — Mechanism: Prolonged glucocorticoid excess. Presentation: Proximal weakness, easy bruising, purple striae.

Investigations: Late-night salivary cortisol, urine free cortisol or dexamethasone suppression. **Interpretation / caution:** Exclude exogenous steroid use; screening tests need validated collection and confirmatory endocrinology assessment.

Primary hyperparathyroidism — Mechanism: Inappropriately high PTH, often adenoma. Presentation: Stones, osteoporosis, sometimes asymptomatic.

Investigations: Corrected/ionized calcium, PTH, phosphate, vitamin D. **Interpretation / caution:** Elevated calcium with nonsuppressed PTH is suggestive; distinguish familial hypocalciuric hypercalcemia.

Osteoporosis — Mechanism: Low bone strength; age, menopause, steroids. Presentation: Fragility fracture or asymptomatic.

Investigations: DXA, fracture history, calcium, renal function, secondary workup. **Interpretation / caution:** T-score <= -2.5 applies to appropriate adult groups; fragility fracture may establish clinical diagnosis.

Cardiovascular

Hypertension — Mechanism: Primary multifactorial; secondary renal/endocrine causes. Presentation: Usually no symptoms.

Investigations: Repeated office BP; home/ambulatory BP; renal tests. **Interpretation / caution:** Use standardized technique and jurisdiction-specific thresholds; exclude white-coat effect.

Acute coronary syndrome — Mechanism: Coronary plaque rupture/erosion or supply-demand imbalance. Presentation: Chest pressure, diaphoresis, nausea, dyspnea.

Investigations: Immediate ECG, serial high-sensitivity troponin. **Interpretation / caution:** MI requires dynamic troponin injury plus ischemic evidence; initial normal ECG/troponin does not exclude early ACS.

Chronic coronary disease — Mechanism: Atherosclerotic coronary narrowing. Presentation: Exertional chest discomfort, reduced exercise tolerance.

Investigations: Clinical probability, ECG, CCTA or functional testing. **Interpretation / caution:** Choose modality based on pretest probability, renal function, exercise capacity and local pathway.

Heart failure — Mechanism: Myocardial or valvular dysfunction. Presentation: Dyspnea, orthopnea, edema, fatigue.

Investigations: BNP/NT-proBNP, ECG, echocardiography. **Interpretation / caution:** Natriuretic peptides vary with age, obesity, kidney function and AF; echo determines phenotype.

Atrial fibrillation — Mechanism: Atrial electrical/structural remodeling. Presentation: Irregular pulse, palpitations, dyspnea, stroke.

Investigations: 12-lead ECG or rhythm monitoring; TSH, electrolytes. **Interpretation / caution:** Diagnosis requires rhythm documentation; assess hemodynamic stability and stroke/bleeding risk.

Peripheral artery disease — Mechanism: Limb atherosclerosis. Presentation: Claudication, wounds, rest pain.

Investigations: Ankle-brachial index, duplex imaging. **Interpretation / caution:** ABI <=0.90 supports PAD; calcified noncompressible vessels may require toe pressures.

Deep vein thrombosis — Mechanism: Venous thrombosis due to stasis, injury or hypercoagulability. Presentation: Unilateral swollen painful leg.

Investigations: Clinical probability, D-dimer, compression ultrasound. **Interpretation / caution:** D-dimer excludes only in appropriate low-risk pathways; positive result is nonspecific.

Pulmonary embolism — Mechanism: Embolization of venous clot. Presentation: Sudden dyspnea, pleuritic pain, tachycardia.

Investigations: Pretest probability, D-dimer, CTPA/VQ scan. **Interpretation / caution:** Unstable patients require immediate emergency pathway; avoid indiscriminate D-dimer testing.

Renal & urinary

Chronic kidney disease — Mechanism: Persistent structural/functional kidney abnormality. Presentation: Often silent; edema, fatigue late.

Investigations: eGFR, urine ACR, urinalysis, imaging if indicated. **Interpretation / caution:** Abnormality generally >=3 months; stage using G and A categories; G1/G2 alone do not establish CKD.

Acute kidney injury — Mechanism: Prerenal, intrinsic or obstructive insult. Presentation: Oliguria, edema or incidental creatinine rise.

Investigations: Serial creatinine, urine output, urinalysis, ultrasound if obstruction. **Interpretation / caution:** KDIGO: creatinine >=0.3 mg/dL rise in 48 h, >=1.5x in 7 days, or urine-output criterion.

Cystitis — Mechanism: Usually bacterial bladder infection. Presentation: Dysuria, urgency, frequency.

Investigations: Symptoms; urinalysis/culture when indicated. **Interpretation / caution:** Asymptomatic bacteriuria is not synonymous with UTI; avoid routine treatment except specified groups.

Pyelonephritis — Mechanism: Ascending kidney infection. Presentation: Fever, flank pain, nausea.

Investigations: Urine culture, urinalysis, renal function. **Interpretation / caution:** Culture guides therapy; image if obstruction, severe illness or nonresponse.

Nephrolithiasis — Mechanism: Urinary crystal precipitation. Presentation: Colicky flank pain, hematuria.

Investigations: Urinalysis, ultrasound or noncontrast CT. **Interpretation / caution:** Fever with obstructing stone is a urologic emergency.

Glomerulonephritis — Mechanism: Immune-mediated glomerular inflammation. Presentation: Hematuria, proteinuria, edema, hypertension.

Investigations: Urine microscopy, ACR/PCR, creatinine, complement, serology. **Interpretation / caution:** RBC casts and dysmorphic RBCs support glomerular bleeding; urgent nephrology if rapidly progressive.

Gastrointestinal & liver

MASLD — Mechanism: Hepatic steatosis with cardiometabolic dysfunction. Presentation: Usually asymptomatic.

Investigations: Liver tests, ultrasound/elastography, FIB-4. **Interpretation / caution:** ALT can be normal; FIB-4 is a fibrosis triage tool, not a standalone diagnosis.

Hepatitis B — Mechanism: HBV infection. Presentation: Often asymptomatic; jaundice, fatigue.

Investigations: HBsAg, anti-HBs, total anti-HBc; HBV DNA. **Interpretation / caution:** Interpret serology as a panel; HBV DNA helps assess viral replication.

Hepatitis C — Mechanism: HCV infection. Presentation: Often silent.

Investigations: Anti-HCV then HCV RNA. **Interpretation / caution:** Antibody indicates exposure; detectable RNA establishes current infection.

Cirrhosis — Mechanism: Advanced hepatic fibrosis. Presentation: Ascites, varices, jaundice, encephalopathy.

Investigations: CBC, INR, albumin, bilirubin, imaging/elastography. **Interpretation / caution:** Normal enzymes do not exclude cirrhosis; assess synthetic function and complications.

Acute pancreatitis — Mechanism: Gallstones, alcohol, high triglycerides, other causes. Presentation: Severe epigastric pain radiating to back, vomiting.

Investigations: Lipase, metabolic panel, ultrasound; imaging selectively. **Interpretation / caution:** Usually 2 of 3: typical pain, enzyme >=3x ULN, characteristic imaging.

Biliary colic/cholecystitis — Mechanism: Gallstones with transient or persistent cystic duct obstruction. Presentation: RUQ pain; fever/tenderness if cholecystitis.

Investigations: Ultrasound, CBC, bilirubin/liver enzymes. **Interpretation / caution:** Ultrasound is first-line; jaundice or sepsis suggests ductal obstruction/cholangitis.

Peptic ulcer disease — Mechanism: H. pylori or NSAID-related mucosal injury. Presentation: Epigastric pain, melena if bleeding.

Investigations: H. pylori stool antigen/breath test; endoscopy when indicated. **Interpretation / caution:** PPI/antibiotics can affect H. pylori testing; bleeding and alarm signs require urgent assessment.

Celiac disease — Mechanism: Gluten-triggered immune enteropathy. Presentation: Diarrhea, bloating, anemia, weight loss.

Investigations: tTG-IgA, total IgA; biopsy by pathway. **Interpretation / caution:** Test while consuming gluten; IgA deficiency changes the test strategy.

Inflammatory bowel disease — Mechanism: Chronic intestinal immune inflammation. Presentation: Diarrhea, blood, weight loss, pain.

Investigations: Fecal calprotectin, CRP, colonoscopy with biopsy. **Interpretation / caution:** Calprotectin is not disease-specific; exclude infection.

Respiratory

Asthma — Mechanism: Variable airway inflammation/obstruction. Presentation: Episodic wheeze, cough, dyspnea.

Investigations: Spirometry with bronchodilator response, peak-flow variability. **Interpretation / caution:** Demonstrate variable expiratory airflow limitation when feasible; normal spirometry between attacks does not exclude.

COPD — Mechanism: Chronic airway/alveolar damage from exposures. Presentation: Chronic dyspnea, cough, sputum.

Investigations: Post-bronchodilator spirometry. **Interpretation / caution:** FEV1/FVC <0.70 supports persistent obstruction; consider age-related limitations.

Pneumonia — Mechanism: Infectious alveolar inflammation. Presentation: Fever, cough, dyspnea, pleuritic pain.

Investigations: Chest imaging, SpO2, CBC, selected microbiology. **Interpretation / caution:** Interpret imaging with clinical syndrome; consider severity and sepsis.

Pulmonary tuberculosis — Mechanism: Mycobacterium tuberculosis. Presentation: Chronic cough, fever, sweats, weight loss.

Investigations: Sputum NAAT and culture, chest imaging. **Interpretation / caution:** TST/IGRA do not distinguish latent from active TB; obtain microbiological confirmation when possible.

Obstructive sleep apnea — Mechanism: Recurrent sleep-related upper airway obstruction. Presentation: Snoring, witnessed apneas, daytime sleepiness.

Investigations: Polysomnography/home sleep apnea testing when appropriate. **Interpretation / caution:** AHI and symptoms/comorbidities determine clinical significance; home tests have limitations.

Hematology

Iron deficiency anemia — Mechanism: Blood loss, low intake, malabsorption. Presentation: Fatigue, pallor, exertional dyspnea.

Investigations: CBC, ferritin, transferrin saturation. **Interpretation / caution:** Low ferritin strongly supports deficiency; inflammation can elevate ferritin and mask deficiency.

Vitamin B12 deficiency — Mechanism: Dietary deficiency, pernicious anemia, malabsorption. Presentation: Macrocytosis, neuropathy, gait problems.

Investigations: CBC, B12, MMA when equivocal. **Interpretation / caution:** Neurologic deficits may occur without anemia; MMA affected by kidney function.

Thalassemia — Mechanism: Inherited globin chain synthesis defect. Presentation: Microcytosis, variable anemia.

Investigations: CBC, iron studies, hemoglobin analysis. **Interpretation / caution:** Microcytosis with normal iron status suggests hemoglobinopathy; do not assume iron deficiency.

Hemolytic anemia — Mechanism: Immune, inherited or mechanical RBC destruction. Presentation: Jaundice, dark urine, anemia.

Investigations: Reticulocytes, LDH, bilirubin, haptoglobin, smear, DAT. **Interpretation / caution:** Pattern supports hemolysis; DAT is helpful for immune etiologies, not all causes.

Acute leukemia — Mechanism: Malignant hematopoietic proliferation. Presentation: Fatigue, bruising, infections, fever.

Investigations: CBC/smear, marrow, flow cytometry, cytogenetics. **Interpretation / caution:** Blasts/cytopenias need urgent hematology review; normal total WBC does not exclude.

Neurology

Ischemic stroke — Mechanism: Acute cerebral vessel occlusion. Presentation: Sudden weakness, aphasia, vision loss.

Investigations: Immediate glucose, CT/MRI, vascular imaging. **Interpretation / caution:** Time of onset and imaging govern reperfusion eligibility; do not delay stroke activation.

Intracranial hemorrhage — Mechanism: Vascular rupture or trauma. Presentation: Sudden severe headache, deficit, decreased consciousness.

Investigations: Emergency noncontrast CT; vascular imaging selectively. **Interpretation / caution:** Urgent neurocritical evaluation; lumbar puncture only in appropriate pathways.

Epilepsy — Mechanism: Predisposition to recurrent unprovoked seizures. Presentation: Seizures, altered awareness.

Investigations: Clinical history, EEG, MRI when indicated. **Interpretation / caution:** Normal routine EEG does not exclude epilepsy; exclude acute symptomatic causes.

Migraine — Mechanism: Primary headache disorder. Presentation: Episodic headache, photophobia, nausea.

Investigations: Clinical diagnostic criteria; imaging if red flags. **Interpretation / caution:** Sudden thunderclap, new focal deficit or systemic illness warrants secondary headache workup.

Peripheral neuropathy — Mechanism: Diabetes, B12 deficiency, toxins and others. Presentation: Distal numbness, burning, weakness.

Investigations: Neurological exam, HbA1c, B12, SPEP selectively, NCS. **Interpretation / caution:** Pattern (axonal/demyelinating, focal/generalized) directs testing.

Rheumatology

Rheumatoid arthritis — Mechanism: Autoimmune synovitis. Presentation: Symmetric inflammatory small-joint pain, stiffness.

Investigations: Exam, RF, anti-CCP, ESR/CRP, imaging. **Interpretation / caution:** Seronegative RA exists; clinical synovitis is central.

Systemic lupus erythematosus — Mechanism: Multisystem autoimmune disease. Presentation: Rash, arthritis, nephritis, cytopenias.

Investigations: ANA, anti-dsDNA, C3/C4, urine protein, CBC. **Interpretation / caution:** Positive ANA alone is not diagnostic; assess organ involvement.

Gout — Mechanism: Monosodium urate crystal deposition. Presentation: Acute red hot swollen joint.

Investigations: Joint aspirate with crystal microscopy, serum urate. **Interpretation / caution:** Crystals confirm; serum urate can be normal during flare; exclude septic arthritis.

Infectious diseases

Sepsis — Mechanism: Dysregulated host response to infection with organ dysfunction. Presentation: Fever or hypothermia, confusion, hypotension.

Investigations: Clinical assessment, lactate, cultures, organ function tests. **Interpretation / caution:** No single biomarker diagnoses sepsis; immediate resuscitation and source evaluation.

HIV — Mechanism: Human immunodeficiency virus infection. Presentation: Acute viral syndrome or asymptomatic.

Investigations: 4th-gen Ag/Ab with reflex algorithm; RNA when indicated. **Interpretation / caution:** Window periods and confirmatory algorithm matter.

Dengue — Mechanism: Arboviral infection. Presentation: Fever, myalgia, headache, rash.

Investigations: NS1/NAAT early; IgM later, CBC/hematocrit. **Interpretation / caution:** Watch for warning signs, plasma leakage and platelet trends; timing affects assay choice.

Malaria — Mechanism: Plasmodium infection. Presentation: Fever, chills, headache, anemia.

Investigations: Thick/thin smears, rapid test. **Interpretation / caution:** A negative initial test may need repetition if suspicion remains; assess severity.

Oncology

Colorectal cancer — Mechanism: Neoplastic transformation, hereditary/sporadic. Presentation: Bleeding, altered bowel habit, iron deficiency anemia.

Investigations: Colonoscopy with biopsy; FIT for screening. **Interpretation / caution:** Positive FIT requires diagnostic colonoscopy; do not use tumor markers as standalone diagnosis.

Breast cancer — Mechanism: Malignant breast epithelial neoplasm. Presentation: Mass, nipple/skin changes.

Investigations: Age-appropriate imaging and core biopsy. **Interpretation / caution:** Histopathology establishes diagnosis; receptor testing guides subsequent care.

Lung cancer — Mechanism: Smoking, radon, occupational and genetic factors. Presentation: Cough, hemoptysis, weight loss.

Investigations: CT, biopsy, histology/molecular testing. **Interpretation / caution:** Low-dose CT is screening in eligible high-risk groups, not a diagnostic substitute.

Cervical cancer — Mechanism: Persistent oncogenic HPV. Presentation: Abnormal bleeding, often silent early.

Investigations: HPV/cytology screening, colposcopy/biopsy. **Interpretation / caution:** Abnormal screening is not a cancer diagnosis; biopsy is diagnostic.

Psychiatry

Major depressive disorder — Mechanism: Multifactorial. Presentation: Low mood/anhedonia, sleep and appetite changes.

Investigations: Clinical interview, PHQ-9 as support. **Interpretation / caution:** Screening score alone is not diagnosis; assess suicidality and bipolar history.

Generalized anxiety disorder — Mechanism: Multifactorial. Presentation: Excessive worry, tension, sleep problems.

Investigations: Clinical interview, GAD-7 as support. **Interpretation / caution:** Exclude medication/substance, thyroid and cardiopulmonary mimics when indicated.

3. Core laboratory interpretation

Always verify patient identity, collection time, specimen, unit, assay-specific reference interval, pregnancy status, medications and prior results. Reference intervals are not the same as disease decision thresholds. Values below are approximate adult teaching ranges and must not replace local laboratory ranges.

Test	Illustrative reference	Clinical interpretation
Hemoglobin	~12–15.5 g/dL (women); ~13.5–17.5 (men)	Low: anemia; evaluate MCV, reticulocytes and blood loss.
WBC	~4.0–11.0 x10^9/L	High/low are nonspecific; assess differential and smear.
Platelets	~150–450 x10^9/L	Low: bleeding/consumption/marrow issues; high: reactive or clonal.
MCV	~80–100 fL	Microcytosis: iron deficiency/thalassemia; macrocytosis: B12/folate, liver, drugs.
Sodium	~135–145 mmol/L	Interpret corrected sodium in hyperglycemia and assess symptoms/chronicity.
Potassium	~3.5–5.0 mmol/L	Critical changes require ECG/urgent review; exclude hemolysis-related pseudohyperkalemia.
Creatinine	Lab and patient dependent	Trend and eGFR; consider muscle mass and AKI.
Calcium	~8.5–10.5 mg/dL total	Check albumin/ionized calcium, PTH and symptoms.
ALT / AST	Lab specific	Hepatocellular injury pattern; AST also from muscle.
ALP / bilirubin	Lab specific	Cholestatic pattern; correlate with GGT and imaging.
Albumin	~3.5–5.0 g/dL	Low with inflammation, hepatic disease, renal loss and others.
TSH / free T4	Assay specific	Interpret together; pituitary disease may invalidate TSH-only approach.
Ferritin / TSAT	Context specific	Low ferritin supports iron deficiency; inflammation can

		elevate ferritin.
Troponin	Assay-specific 99th percentile	Elevated = myocardial injury; MI additionally requires ischemic evidence.
D-dimer	Assay and unit specific	Only use with validated pretest probability pathway.
CRP / ESR	Assay/age dependent	Nonspecific inflammation; neither rules infection in/out alone.
BNP / NT-proBNP	Age/setting dependent	Renal function, obesity and AF modify interpretation.
HbA1c	<5.7% normoglycemic ADA range	Interference with altered RBC lifespan/hemoglobin variants.

Focused diagnostic thresholds

- Diabetes (nonpregnant adults): HbA1c >=6.5%, FPG >=126 mg/dL, 2-hour 75-g OGTT >=200 mg/dL, or random plasma glucose >=200 mg/dL with classic symptoms/hyperglycemic crisis. Confirmation rules apply.
- CKD: abnormalities of kidney structure or function for >=3 months. eGFR G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73m2. ACR A1 <30, A2 30–300, A3 >300 mg/g.
- AKI: creatinine rise >=0.3 mg/dL in 48 hours, >=1.5x baseline in 7 days, or qualifying oliguria.
- Acute pancreatitis: 2 of typical pain, lipase/amylase >=3x upper limit, or imaging findings.
- COPD: post-bronchodilator FEV1/FVC <0.70 in compatible clinical context.
- Iron deficiency: ferritin threshold depends on inflammation and population; very low ferritin is strongly suggestive.
- Troponin: myocardial injury above assay 99th percentile; dynamic change plus ischemic evidence required for MI.

4. Symptom-based diagnostic pathways

Chest pain

Approach: First exclude ACS, aortic dissection, PE, tension pneumothorax; ECG and urgent risk-stratified testing.

Teaching caution: Avoid labeling as reflux before time-critical causes are considered.

Dyspnea

Approach: Assess SpO2, respiratory effort, vitals; consider asthma/COPD, HF, pneumonia, PE, anemia, metabolic acidosis.

Teaching caution: Normal resting SpO2 does not exclude PE or serious illness.

Fatigue

Approach: Characterize duration, sleep, mood, medication, bleeding, weight change; CBC, glucose, TSH and targeted studies as indicated.

Teaching caution: Do not order indiscriminate broad panels without clinical direction.

Polyuria/polydipsia

Approach: Check plasma glucose and urinalysis; consider diabetes, diuretics, hypercalcemia, diabetes insipidus.

Teaching caution: Differentiate urinary frequency from true high urine volume.

Fever

Approach: Assess instability, exposures, focal symptoms; targeted cultures, imaging and region-specific infection tests.

Teaching caution: Normal WBC or CRP does not exclude serious infection.

Abdominal pain

Approach: Localize pain, assess peritonitis, bleeding, obstruction, vascular and pregnancy-related causes.

Teaching caution: Pregnancy test when biologically relevant; urgent imaging if red flags.

Headache

Approach: Assess thunderclap onset, fever/meningism, focal deficits, immunosuppression and new pattern.

Teaching caution: Do not classify as migraine before excluding red flags.

Leg edema

Approach: Assess unilateral vs bilateral; DVT, HF, renal/liver disease, medication and venous causes.

Teaching caution: Sudden unilateral swelling needs VTE risk evaluation.

Anemia

Approach: Confirm CBC and indices; reticulocytes, ferritin/iron, B12, folate, hemolysis studies and bleeding evaluation by pattern.

Teaching caution: Identify source, not only deficiency.

Weight loss

Approach: Confirm trajectory; consider malignancy, diabetes, thyroid, infection, malabsorption, mood and social factors.

Teaching caution: Investigate alarm symptoms and examination findings.

5. Diagnostic pitfalls and safe handover

- Do not confuse a screening result with definitive diagnosis (e.g., positive FIT, ANA, D-dimer or PSA).
- Do not interpret laboratory numbers without units, assay-specific limits, age/pregnancy context and trends.
- Do not assume elevated troponin always indicates type 1 MI, or that normal early troponin excludes ACS.
- Do not use a single eGFR to diagnose chronic kidney disease without chronicity or other evidence.
- Do not interpret positive urine culture as symptomatic UTI without symptoms/context.
- Do not infer active tuberculosis from a positive IGRA or TST alone.
- Do not let a negative test override high pretest probability without understanding test sensitivity and timing.
- Document working diagnosis, alternatives, pending tests, uncertainty, follow-up and return precautions.

Structured clinical handover (SBAR)

- Situation: age, presenting problem, acuity, current vital signs.
- Background: relevant comorbidities, medications, allergies, recent events.
- Assessment: examination, key positive/negative findings, leading diagnosis and dangerous alternatives.
- Recommendation: immediate actions, tests pending, specialist escalation, reassessment and disposition.

6. Recommended authoritative references

- ADA Standards of Care in Diabetes — https://diabetesjournals.org/care
- KDIGO Clinical Practice Guidelines — https://kdigo.org/guidelines/
- NICE Guidance — https://www.nice.org.uk/guidance
- European Society of Cardiology Guidelines — https://www.escardio.org/Guidelines
- AHA/ACC Clinical Practice Guidelines — https://professional.heart.org/en/guidelines-and-statements
- GOLD COPD Report — https://goldcopd.org/
- GINA Asthma Reports — https://ginasthma.org/

- WHO TB Guidelines — https://www.who.int/teams/global-tuberculosis-programme
- Surviving Sepsis Campaign — https://www.sccm.org/clinical-resources/guidelines
- WHO ICD-11 — https://icd.who.int/
- LOINC — https://loinc.org/
- SNOMED CT — https://www.snomed.org/

Reference policy: This draft identifies source organizations and teaching-standard criteria but does not claim that each statement has been checked against the latest guideline revision. Prior to clinical deployment, assign a named clinical reviewer, validate every threshold against the current source edition, record jurisdiction and effective date, and maintain a change log.